



JAIPUR NATIONAL UNIVERSITY

Semester/Year End (Main/Back) Examinations, May-2025

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science Engineering)
Semester/Year:	IV / II
Paper Code:	BTECSDS401T22
Title of Paper:	Communication Fundamentals
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION-A	Marks Allotted
Q.1	Define the primary and secondary line constant?	4
Q.2	The antenna current of an AM transmitter is 10 Ampere when it is modulated to the depth of 30% by an audio signal. It increases to 11 ampere when another signal modulates the carrier signal. What will be the modulation index due to second signal?	4
Q.3	Prove that the balance modulator produces an output consisting of sideband only with the carrier removed?	4
Q.4	Draw the block diagram of narrowband FM generation and explain it?	4
Q.5	State and prove sampling theorem.	4
Q.6	What is quantization? Explain uniform quantization.	4
Q.7	Explain Bluetooth technology in brief?	4

SECTION-B

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| Q.1 | Explain all types of intrinsic and extrinsic losses in optical fiber? | 10 |
| Q.2 | Explain Square law diode modulator for AM generation? | 10 |
| Q.3 | Explain the phase shift method of SSB generation. List its advantages and disadvantages with respect to the other methods. | 10 |
| Q.4 | With the help of neat diagram, explain the transmitter and receiver of pulse code modulation. | 10 |
| Q.5 | Explain all types of analog pulse modulation methods with the help of waveforms. | 10 |
| Q.6 | Write short note on the following:
(a) Wi-MAX
(b) GPRS | 10 |